6/05

```
e3 = e1+e2;
e4 = e1-10
e5 = --e1;
e6 = e2--;
```

```
Complex e2=e1;
Complex e3=e1+e2;
```

copy constructor

```
Complex e2;
e2=e1;
Complex e3;
e3=e1+e2;
```

Assignment operator

If a programmer does not supply the assignment operator overloaded function, compiler will supply

.

As long as data members are statically created ones we can happily depend on COmpiler as assignment operator overloaded function.

If data members are dynamically allocated we need to supply assignment operator over loaded function with deep copy.

What are the functions suplied by the compiler?

1. default constructor
2. copy constructor
3. destructor
4. assignment operator overloaded function

operator overloading using friend functions

```
e3=e1+e2;
e4=e1+10;
e5=--e1;
e6=e2++;
```

Can implemented with member functions

```
e3=15+e1;
```

✗ Cannot be implemented with member functions because of Left operand is not Correct object

The only solution we have is friend function, In friend function we need to pass both left operand and right operand as a parameters

```
e3=e1+e2;      operator+(e1,e2);
e4=e2-e1;      operator-(e2,e1);
e5=e3+10;      operator+(e3,10);
e6=20-e4;      operator-(20,e4);
e7=--e5;       operator--(e5);
e8=e6++;       operator++(e6,int);
```

Operators that cannot be overloaded by friend functions

1. sizeof
2.scope resolution
3. ternary
4.typeid
5. dot
6. *.
7. =
8. ->
9. []
10 .()

String — char *ptr;
int size

Constructor
Copy cons
Destructor
Assignment operator ovl
+
==
[]